

Partha Ghosh

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Partha Portfolio | ppaartha | ppaartha

I build LLM agents and production ML that sit on real systems, not demo chatbots. At ConnectedAgents I work on [Optiify](#) (LangGraph tool-calling, MCP live point data, Neo4j graphs, and RAG) and on [Quantiify](#) (YOLOv8 + SAHI for drawing symbol count and legend detection). Previously at REVE Systems I shipped RAG chat for [REVE Chat](#) using GraphRAG, LightRAG, multimodal RAG, agentic RAG, and LangChain, Bengali STT/TTS ([Kotha](#), [Uccharon](#)), Temporal Fusion Transformer forecasts and a CatVTON virtual try-on for [Le Reve](#).

EDUCATION

Shahjalal University of Science and Technology

Sylhet, Bangladesh

B.Sc. in Computer Science and Engineering
CGPA: 3.35 / 4.00 (Last 4 semesters: 3.61)

Feb 2018 – Mar 2023

WORK EXPERIENCE

ConnectedAgents AI Engineer

Dec 2025 – Present

- Built a multi-agent system using **LangGraph** for [Optiify](#) copilot so building managers can automate after-hours HVAC governance, enabling plain-language tenant self-scheduling, automated usage logging and cost-leak prevention alongside real-time fault inspection of HVAC systems.
- Designed tool-calling workflows (booking, FDD, work orders, building context) with **FastAPI**, **DynamoDB** and **InfluxDB** instead of stuffing raw building data into the prompt.
- Built an automated **evaluation** pipeline with test cases to benchmark agent tool routing precision, argument extraction and multi-turn workflows across FDD, booking and RAG tools.
- Implemented an **MCP** agent for live point-condition time series: catalog lookup in **DynamoDB**, per-point **Influx** queries and prompt-led date windows.
- Built a **Neo4j** knowledge graph of HVAC equipment hierarchy, with **APOC**/full-text tools so the agent can search and traverse.
- Added document **RAG** over building functional docs and onboarding FAQs to ground fault explanations in site-specific knowledge.
- Trained **YOLOv8** with **SAHI** for [Quantiify](#) to detect legends and count symbols on large construction drawings without missing small objects.

REVE Systems Machine Learning Engineer

Jan 2024 – Dec 2025

- Built a **RAG** chatbot for [REVE Chat](#) using **GraphRAG**, **LightRAG**, multimodal **RAG**, agentic **RAG**, and **LangChain**, with hybrid search (**BM25** + dense) for stronger responses.
- Fine-tuned **wav2vec 2.0** and **Whisper** for Bengali **STT** ([Kotha](#)) with **PEFT-LoRA**; evaluated on multiple test sets, including augmented data, to find model weaknesses.
- Fine-tuned a **VITS** model for Bengali **TTS** ([Uccharon](#)) with custom **G2P** phonemization for accurate pronunciation and natural prosody.
- Developed a voice-based banking agent with an **STT**-chatbot-**TTS** pipeline to automate customer conversations.
- Compressed models with quantization and **ONNX** for faster load/inference and offline use.
- Implemented face liveness detection for the NTMC biometric authentication project using **OpenCV**, **CNNs**, and **MediaPipe** (anti-spoofing).
- Developed a **Temporal Fusion Transformer (TFT)** pipeline for [Le Reve](#) to forecast production quantity and sales over multiple horizons, with attention-based interpretability so planners could see which product, season, and store signals drove each prediction.
- Built a virtual try-on for [Le Reve](#) with **CatVTON**, a lightweight diffusion model, so customers could see garments on a person image without a heavy try-on stack.
- Developed a chatbot with **BERT** and **Bi-LSTM**, and improved the [REVE Chat](#) dashboard frontend (**React.js**).

Codelab FZC Software Engineer

Jun 2023 – Jan 2024

- Developed a 2D object-detection game and a Japanese typing game in **Unity** for Android and iOS.
- Contributed to an admin panel, implementing CRUD operations in **Laravel**.

TECHNICAL SKILLS & TECHNOLOGIES

Languages	Python, Java, C, C++, PHP, C#
Agentic AI	LangGraph, LangChain, LlamaIndex, MCP, Agentic RAG, OpenAI, Groq
Machine Learning	PyTorch, TensorFlow, OpenCV, Scikit-learn, MLflow, CI/CD
Web Technologies	Django, FastAPI, JavaScript (React.js, Node.js, Vue.js), HTML/CSS, Laravel
Databases	MySQL, PostgreSQL, MongoDB, Qdrant, Memgraph, Neo4j, InfluxDB, DynamoDB, Firebase
Cloud / Infra	AWS, Docker, Linux, Git, Bash
Tools	L ^A T _E X, Microsoft Office
Problem Solving	600+ problems — Codeforces (partha361), LeetCode (ppartha)

RESEARCH EXPERIENCE

- **Development of a Large Corpus of Spontaneous Speech for Improved Speech Recognition in Bangladeshi Bangla:** Created a 175-hour Bangladeshi Bangla spontaneous speech corpus and fine-tuned multilingual **wav2vec 2.0**, comparing it with a reading corpus (**SUBAK.KO**) to improve **ASR**. Conducted under the supervision of [Dr. Mohammad Shahidur Rahman](#)
- **BanSpeech: A Multi-Domain Bangla Speech Recognition Benchmark Toward Robust Performance in Challenging Conditions:** Assessed a Bangla **ASR** model on **BanSpeech**, a 7.2-hour benchmark; it struggled with spontaneous speech and high OOV rates, highlighting domain selection. Published in *IEEE Access*, vol. 12, pp. 34527–34538, 2024. Conducted under the supervision of [Dr. Mohammad Shahidur Rahman](#).

PROJECTS

SUST Student Management System



Web app for student and faculty records, courses, and results, with admin CRUD and faculty mark entry.

Technologies: Django, Python, HTML, CSS, Bootstrap, MySQL

Store Front MCP



MCP server for customers, orders and sales, with an OpenAI assistant that calls tools at runtime.

Technologies: Python, MCP, FastMCP, OpenAI, PostgreSQL, Docker, Pydantic

Automatic Attendance System Using Face Recognition



Streamlit webcam app that recognizes faces and marks attendance from five enrolled photos per classmate.

Technologies: Python, OpenCV, Streamlit

Shoesly - E-commerce Website



Shoe store with User, Supplier, Admin, and Bank routes for product filters, checkout, and order fulfillment.

Technologies: Node.js, React.js, MongoDB, Express.js

ACHIEVEMENTS

- **1st Runner-up** AI for Bangla 2.0 (2023).

EXTRA-CURRICULAR ACTIVITIES

CSE Society, SUST

General Secretary, Assistant General Secretary, Executive Member

2019 – 2023

SUST ACM Student Chapter

Joint Secretary

2021 – 2023